

unu_merit_final_assignment

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1 Final Assignment and Groups

1.1 Network Analysis and Economic Complexity for Sustainable Development

Group 4

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1.2 Python boilerplate

```
[1]: import networkx as nx
import numpy as np
import matplotlib.pyplot as plt
import warnings
warnings.simplefilter(action='ignore', category=FutureWarning)

class ECI:
    def __init__(self, countries, activities, matrix):
        self.countries = countries
        self.activities = activities
        self.matrix = matrix
        self.Mcp = np.array(matrix)
        self.G = self.create_graph()
        self.eci_index = []

    def diversification(self, Mcp=None):
        """Diversification: first order approximation of complexity"""

        if Mcp is None:
            Mcp = self.Mcp

        return Mcp.sum(axis = 1)
```

```

def ubiquity(self, Mcp=None):
    """Ubiquity: first order approximation of complexity"""

    if Mcp is None:
        Mcp = self.Mcp

    return Mcp.sum(axis = 0)

def get_countries_by_diversification(self):
    """Return index and name of countries sorted by diversification"""

    # Countries ordered with respect to their diversification
    order = np.argsort(self.diversification())[::-1]
    names = [self.countries[country] for country in order]

    return order, names

def get_activities_by_ubiquity(self):
    """Return index and name of activities sorted by ubiquity"""

    order = np.argsort(self.ubiquity())[::-1]
    names = [self.activities[activity] for activity in order]

    return order, names

def create_graph(self):
    """Create directed graph"""

    self.R, self.C = self.Mcp.shape

    G = nx.Graph()

    # Add country nodes
    G.add_nodes_from(self.countries, bipartite=0)

    # Add product nodes
    G.add_nodes_from(self.activities, bipartite=1)

    # Add edges where adjacency matrix is 1
    for i in range(self.R):
        for j in range(self.C):
            if self.Mcp[i, j] == 1:
                G.add_edge(self.countries[i], self.activities[j])

    return G

```

```

def show_graph(self, sizex=6, sizey=6):
    """Show directed graph"""

    # Positioning using bipartite layout
    pos = nx.bipartite_layout(self.G, self.countries)

    # Plot the graph
    fig, ax = plt.subplots(1, 1, figsize=(sizex, sizey))
    node_colors = ['lightblue' if node in self.countries else 'darkblue']
    for node in self.G.nodes():

        nx.draw_networkx(self.G, pos, edge_color='lightgray',
        node_color=node_colors, node_size=2500, font_size=12, font_color='grey')

    # Give more space for the names
    x1, x2 = ax.get_xlim()
    plt.xlim([x1 - 0.2, x2])

    plt.title("Directed graph")

    plt.show()

def show_matrix(self, sizex=6, sizey=6):
    """Show an adjacency Matrix"""

    # plt.figure(figsize=(sizex, sizey))
    plt.pcolor(self.Mcp, cmap='Blues')

    plt.gca().invert_yaxis() # Flip the y axis to have the first row on top.
    plt.yticks([i + 0.5 for i in range(len(countries))], countries,
    rotation=45)

    plt.title("Original matrix")

    plt.show()

def show_triangle(self, sizex=6, sizey=6):
    """Show a sorted adjacency Matrix"""

    # plt.figure(figsize=(sizex, sizey))
    #_, ax = plt.subplots()

    # Countries ordered with respect to their diversification
    order, names = self.get_countries_by_diversification()
    triangle = []

```

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        for i in order:
            triangle.append(self.matrix[i])

        plt.pcolor(triangle, cmap='Blues')

        plt.gca().invert_yaxis() # Flip the y axis to have the first row on top.
        plt.yticks([i+0.5 for i in range(len(self.countries))], names,
        ↪rotation=45)

        #ax.set_title("Countries sorted by diversification")
        plt.title("Countries sorted by diversification")

        plt.show()

    def show_diversification(self):
        """Show highlighted diversification - to compare with my triangle"""

        #diversification = np.sum(Mcp, axis = 1)
        #order_countries = np.argsort(diversification)[::-1]
        #
        #ubiquity = np.sum(Mcp, axis = 0)
        #order_products = np.argsort(ubiquity)[::-1]

        order_countries, _ = self.get_countries_by_diversification()
        order_products, _ = self.get_activities_by_ubiquity()

        ordered_Mcp = np.zeros(np.shape(self.Mcp))

        for c in range(len(order_countries)):
            for p in range(len(order_products)):
                ordered_Mcp[c,p] = self.
        ↪Mcp[order_countries[c],order_products[p]]

        plt.matshow(ordered_Mcp, cmap = 'Blues')
        plt.yticks(range(self.R), [self.countries[i] for i in order_countries])
        plt.title("This is lecture chart - activities must have been shift_
        ↪around")

        plt.show()

    def show_ubiquity(self):
        """Show bar chart of most common activities"""

        ubiquity = self.ubiquity()
        order, names = self.get_activities_by_ubiquity()
        count = []
        for i in order:

```

```

        count.append(ubiquity[i])

plt.bar(names, count)
plt.title("Ubiquity of Activities")
plt.ylabel("How many countries")
plt.show()

def show_eci(self, unsign=False, sizex=6, sizey=6):
    """Show ECI indices for the countries"""

    if len(self.eci_index) <= 0:
        raise ValueError("No method was used to set ECI indices")

    if unsign:
        eci = [abs(i) for i in self.eci_index]
    else:
        eci = self.eci_index

    plt.figure(figsize=(sizex, sizey))
    plt.scatter(self.countries, eci)
    plt.xticks([i for i in range(len(self.countries))], self.countries,
↪rotation=45)
    plt.ylabel("ECI")
    plt.title("Scatter plot of ECI values")
    plt.show()

def print(self):
    """Print initial summary"""

    diversification = self.diversification()

    print(f"Original countries:")
    for i in range(len(self.countries)):
        print(f"\tindex '{i:2}' - {self.countries[i]:20} (sum:
↪{diversification[i]})")

    print(f"List of activities:")
    for i in range(len(self.activities)):
        print(f"\tindex '{i:2}' - {self.activities[i]}")

    print(f"Matrix:")
    for line in self.matrix:
        print(f"\t{line}")

    print("\n--- --- ---\n")

def print_diversification(self):

```

```

        """Print sorted countries by diversification"""

        diversification = self.diversification()
        order, names = self.get_countries_by_diversification()

        print(f"Countries sorted by diversification (Most diverse at the top):")
        for pos, i in enumerate(order):
            print(f"\t{pos+1}. - {names[pos]:20} (diversification:␣
↪{diversification[i]})")

        print("--- --- ---")

    def print_ubiquity(self):
        """Print sorted activities by ubiquity"""

        ubiquity = self.ubiquity()
        order, names = self.get_activities_by_ubiquity()

        print(f"Activities sorted by ubiquity (Most common at the top):")
        for pos, i in enumerate(order):
            print(f"\t{pos+1}. - {names[pos]:5} (ubiquity: {ubiquity[i]})")

        print("--- --- ---")

    def OneStep(self, kc=None, kp=None):
        """Function to compute kc(N+1) and kp(N+1) given Mcp and the previous␣
↪step, kc(N) and kp(N)"""

        diversification = self.diversification()
        ubiquity = self.ubiquity()

        if kc is None:
            kc = diversification

        if kp is None:
            kp = ubiquity

        kc1 = self.Mcp.dot(kp)/diversification
        kp1 = self.Mcp.transpose().dot(kc)/ubiquity

        return kc1, kp1

    def MethodOfReflections(self, max_iterations=10):
        """Function to initialize kc(0) and kp(0), iterate OneStep␣
↪max_iterations times, and normalize the output"""

```

```

diversification = self.diversification()
ubiquity = self.ubiquity()

# Setting the initial condition
kc = diversification / diversification.sum()
kp = ubiquity / ubiquity.sum()

for iteration in range(max_iterations):
    kc, kp = self.OneStep(kc, kp)

return (kc - kc.mean())/kc.std(), (kp - kp.mean())/kp.std()

def _print_indices(self, indices):
    for country_index, i in enumerate(indices):
        print(f"\t{country_index:2}: {self.countries[country_index]:20} =_{i:.5}")

def _print_product_indices(self, indices):
    for product_index, i in enumerate(indices):
        print(f"\t{product_index:2}: {self.activities[product_index]:2} =_{i:.5}")

def _print_sorted_indices(self, indices, unsign=False):
    if unsign:
        indices = [abs(i) for i in indices]

    order_countries = np.argsort(indices)[::-1]

    for pos, i in enumerate(order_countries):
        print(f"\t{pos+1}. - ({i:1}) {self.countries[i]:20} = {indices[i]:.5}")

def _print_sorted_product_indices(self, indices, unsign=False):
    if unsign:
        indices = [abs(i) for i in indices]

    order_products = np.argsort(indices)[::-1]

    for pos, i in enumerate(order_products):
        print(f"\t{pos+1}. - ({i:1}) {self.activities[i]:2} = {indices[i]:.5}")

def _save_indices(self, indices, unsign=False):
    self.eci_index = []
    for i in indices:
        if unsign:
            self.eci_index.append(abs(i))

```

```

        else:
            self.eci_index.append(i)

def do_method_of_reflections_once(self):
    """Compute one step, starting from diversification and ubiquity"""

    kc, _ = self.OneStep()
    print("Method of Reflections (One Step only):")
    self._print_indices(kc)
    print("----")

def do_method_of_reflections(self, iterations=10):
    """Compute ECI with the MethodOfReflections"""

    kc, _ = self.MethodOfReflections(iterations)
    print(f"Method of Reflections (Iterations: {iterations}):")
    self._print_indices(kc)
    print("----")

    # store the index for chart
    #self.eci_index = kc
    self._save_indices(kc)

    #Print the order
    print("Ordered:")
    self._print_sorted_indices(kc)
    print("Ordered (sign removed):")
    self._print_sorted_indices(kc, True)

    print("----")

def do_eigen_values_eci(self):
    """ECI as an eigenvalue problem"""

    Mcp = self.Mcp

    # Projecting Matrices
    Ccc = np.matmul(np.array(Mcp.transpose()/np.sum(Mcp,1)).transpose(), np.
↪array(Mcp/np.sum(Mcp,0)).transpose())

    # Compute eigenvalues and eigenvectors
    eigenvalues, eigenvectors = np.linalg.eig(Ccc)

    # ECI is the eigenvector corresponding to the second largest eigenvalue
    ECI = eigenvectors[:,eigenvalues.argsort()[-2]]

    # Normalizing ECI

```



```

ECI = (ECI - ECI.mean())/ECI.std()

print(f"ECI as an eigenvalue problem:")
self._print_indices(ECI)
print("----")

# store the index for chart
self._save_indices(ECI)

#Print the order
print("Ordered:")
self._print_sorted_indices(ECI)
print("Ordered (sign removed):")
self._print_sorted_indices(ECI, True)

print("---- ----")

def do_eigen_values_pci(self):
    """PCI as an eigenvalue problem"""

    Mcp = self.Mcp

    # Projecting Matrices
    Cpp = np.matmul(np.array(Mcp/np.sum(Mcp,0)).transpose(),np.array(Mcp.
↪transpose()/np.sum(Mcp,1)).transpose())

    # Compute eigenvalues and eigenvectors
    eigvalues, eigvectors = np.linalg.eig(Cpp)

    # PCI is the eigenvector corresponding to the second largest eigenvalue
    PCI = eigvectors[:,eigvalues.argsort()[-2]]

    # Normalizing PCI
    PCI = (PCI - PCI.mean())/PCI.std()

    print(f"PCI as an eigenvalue problem:")
    self._print_product_indices(PCI)
    print("----")

    #Print the order
    print("Ordered:")
    self._print_sorted_product_indices(PCI)
    print("Ordered (sign removed):")
    self._print_sorted_product_indices(PCI, True)

    print("---- ----")

```

```

def fitCompOneStep(self, Fitness):
    """Algorithm to compute Fitness and Complexity"""

    C = self.C

    #One step of the algorithm
    Complexity=np.zeros(C)
    for p in range(C):
        try:
            sumCT=self.Mcp[:,p]/Fitness
            sumCT[np.where(np.isnan(sumCT))]=0.
            sumCT=sumCT*(1-np.isnan(sumCT))
            Complexity[p] = 1./np.sum(sumCT)
            if np.isnan(Complexity[p]):
                Complexity[p]=0
        except ZeroDivisionError:
            Complexity[p] = 0
    Fitness = np.dot(self.Mcp, Complexity)
    return Fitness/np.mean(Fitness),Complexity/np.mean(Complexity)

def FitnessComplexity(self, max_iterations=10):

    diversification = self.diversification()
    ubiquity = self.ubiquity()

    # Setting the initial condition
    F = diversification / diversification.sum()
    Q = ubiquity / ubiquity.sum()

    for iteration in range(max_iterations):
        F,Q = self.fitCompOneStep(F)

    return (F / F.mean()), (Q/Q.mean())

def show_fitness(self, iterations=10):
    N = iterations
    Fseries=[]
    for iteration in range(N):
        F, Q = self.FitnessComplexity(iteration)
        Fseries.append(F)
    Fseries=np.array(Fseries)

    fig,ax=plt.subplots(1,1,)
    plt.plot(Fseries)
    plt.xlabel('Iteration')
    plt.ylabel('Fitness')
    plt.xticks(range(0,N,5),range(0,N,5))

```

```

plt.title("Chart of Fitness")
plt.show()

def show_fitness_and_complexity(self, iterations=10):
    """Fitness and Complexity"""

    F, Q = self.FitnessComplexity(iterations)
    order_countries = np.argsort(F)[::-1]

    ubiquity = self.ubiquity()
    order_products = np.argsort(Q)

    ordered_Mcp = np.zeros(np.shape(self.Mcp))

    for c in range(len(order_countries)):
        for p in range(len(order_products)):
            ordered_Mcp[c,p] = self.
↪Mcp[order_countries[c],order_products[p]]

    plt.matshow(ordered_Mcp, cmap = 'Blues')
    plt.yticks(range(self.R), [countries[i] for i in order_countries],
↪rotation=45)
    plt.xticks(range(self.C), [activities[i] for i in order_products])

    plt.title("Fitness and Complexity")
    plt.show()

def calc_cooccurencies(self):
    self.Jpp = self.Mcp.transpose().dot(self.Mcp)
    return self.Jpp

def print_cooccurencies(self):
    self.calc_cooccurencies()
    print("Relatedness / Co-occurencies")
    print(self.Jpp)
    print("--- --- ---")

def ProductSpace(self):
    Jpp = self.calc_cooccurencies()
    ubiquity = self.ubiquity()

    ubiMat = np.tile(ubiquity, [self.Mcp.shape[1], 1])
    ubiMax = np.maximum(ubiMat, np.transpose(ubiMat)).astype(float)

    np.divide(np.ones_like(ubiMax, dtype=float), ubiMax, out=ubiMax,
↪where=ubiMax != 0)
    return np.multiply(Jpp, ubiMax)

```

```

def Taxonomy(self):
    diversification = self.diversification()
    ubiquity = self.ubiquity()
    Mcp = self.Mcp

    Jpp = Mcp.transpose().dot(np.diag(1/diversification).dot(Mcp))

    ubiMat = np.tile(ubiquity, [Mcp.shape[1],1])
    ubiMax = np.maximum(ubiMat,np.transpose(ubiMat)).astype(float)

    np.divide(np.ones_like(ubiMax,dtype=float), ubiMax, out=ubiMax,
↪where=ubiMax != 0)
    return np.multiply(Jpp,ubiMax)

def AssistMatrix(self):
    diversification = self.diversification()
    ubiquity = self.ubiquity()
    Mcp = self.Mcp

    Jpp = Mcp.transpose().dot(np.diag(1/diversification).dot(Mcp))

    ubiMat = np.tile(ubiquity, [Mcp.shape[1],1]).astype('float')

    np.divide(np.ones_like(ubiMat,dtype=float), ubiMat, out=ubiMat,
↪where=ubiMat != 0)

    return np.multiply(Jpp,ubiMat)

def print_relatedness(self, count=4):
    """Print the relatedness among the first activities/products"""

    print(f"Relatedness / Product Space ({count} products)")
    print(self.ProductSpace()[ :count, :count])
    print("----")

    print(f"Relatedness / Taxonomy ({count} products)")
    print(self.Taxonomy()[ :count, :count])
    print("----")

    # Notice how the Assist Matrix is not symmetric
    print(f"Relatedness / Assist Matrix ({count} products)")
    print(self.AssistMatrix()[ :count, :count])
    print("--- --- ---")

def show_relatedness(self, threshold=0.1):
    Bpp = self.Taxonomy()

```

```

C = self.C

G = nx.Graph()
G.add_nodes_from(self.activities)
G.add_edges_from([ ('P'+str(i), 'P'+str(j)) for i in range(C) for j in
↪range(C) if Bpp[i,j]>threshold])

    nx.draw_networkx(G, pos=nx.spring_layout(G,k=2/np.sqrt(C)),
↪edge_color='lightgray', node_color='lightblue',
↪node_size=2500,font_size=12,font_color='grey')

plt.show()

def show_proximity_heatmap(self, sizex=8, sizey=6):
    proximity = self.Taxonomy()

    plt.figure(figsize=(sizex, sizey))
    plt.imshow(proximity, cmap='RdYlBu_r', aspect='auto',
↪interpolation='nearest')
    plt.colorbar(label='Relatedness (Proximity)')
    plt.xticks(range(len(activities)), activities, rotation=45, ha='right')
    plt.yticks(range(len(activities)), activities)
    plt.title('Product Relatedness Heatmap', fontsize=14)
    plt.tight_layout()
    plt.show()

def show_activity_heatmap(self, sizex=6, sizey=5):
    Jpp = self.calc_cooccurencies()

    plt.figure(figsize=(sizex, sizey))
    plt.pcolor(Jpp, cmap='Blues')
    plt.gca().invert_yaxis()
    plt.xticks([i + 0.5 for i in range(len(self.activities))], self.
↪activities)
    plt.yticks([i + 0.5 for i in range(len(self.activities))], self.
↪activities)
    plt.title("Relatedness (Co-occurencies)")
    plt.xlabel("Activities")
    plt.ylabel("Activities")
    plt.colorbar(label="Activity co-occurency")
    plt.tight_layout()
    plt.show()

```

1.3 Main section

1.3.1 Initialization and Data

```
[2]: # Define matrix, connecting (imaginary) countries and activities.

countries=['Vale of Arryn','The North','The Riverlands', 'Dorne', 'The_
↳Stormland','The Westerlands','The Reach']
activities=[f'P{j}' for j in range(9)]

matrix = [[1, 0, 1, 0, 1, 1, 0, 0, 1],
          [0, 1, 1, 0, 0, 1, 1, 0, 0],
          [0, 1, 0, 1, 1, 0, 0, 0, 0],
          [1, 0, 1, 0, 0, 0, 0, 0, 0],
          [0, 0, 0, 0, 0, 1, 1, 0, 1],
          [0, 1, 1, 0, 1, 0, 1, 1, 1],
          [1, 1, 1, 1, 1, 0, 1, 1, 1]]
```

```
[3]: Index1 = ECI(countries, activities, matrix)
Index1.print()
```

Original countries:

index ' 0' - Vale of Arryn	(sum: 5)
index ' 1' - The North	(sum: 4)
index ' 2' - The Riverlands	(sum: 3)
index ' 3' - Dorne	(sum: 2)
index ' 4' - The Stormland	(sum: 3)
index ' 5' - The Westerlands	(sum: 6)
index ' 6' - The Reach	(sum: 8)

List of activities:

index ' 0' - P0
index ' 1' - P1
index ' 2' - P2
index ' 3' - P3
index ' 4' - P4
index ' 5' - P5
index ' 6' - P6
index ' 7' - P7
index ' 8' - P8

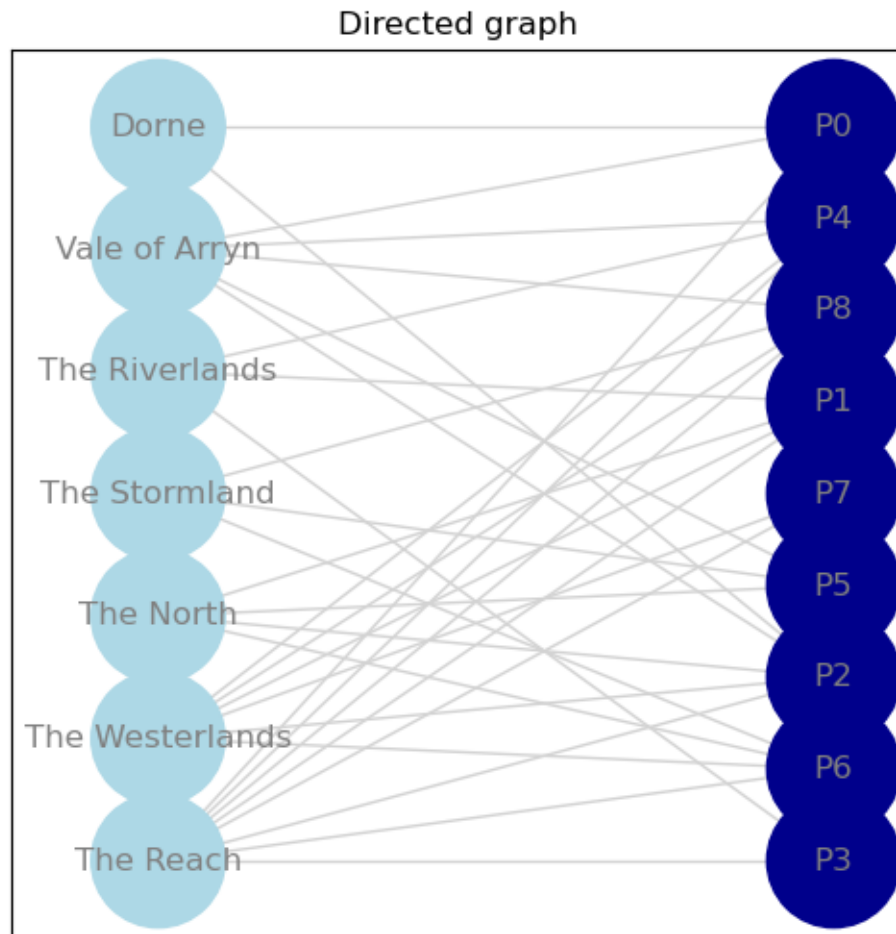
Matrix:

[1, 0, 1, 0, 1, 1, 0, 0, 1]
[0, 1, 1, 0, 0, 1, 1, 0, 0]
[0, 1, 0, 1, 1, 0, 0, 0, 0]
[1, 0, 1, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 1, 1, 0, 1]
[0, 1, 1, 0, 1, 0, 1, 1, 1]
[1, 1, 1, 1, 1, 0, 1, 1, 1]

--- --- ---

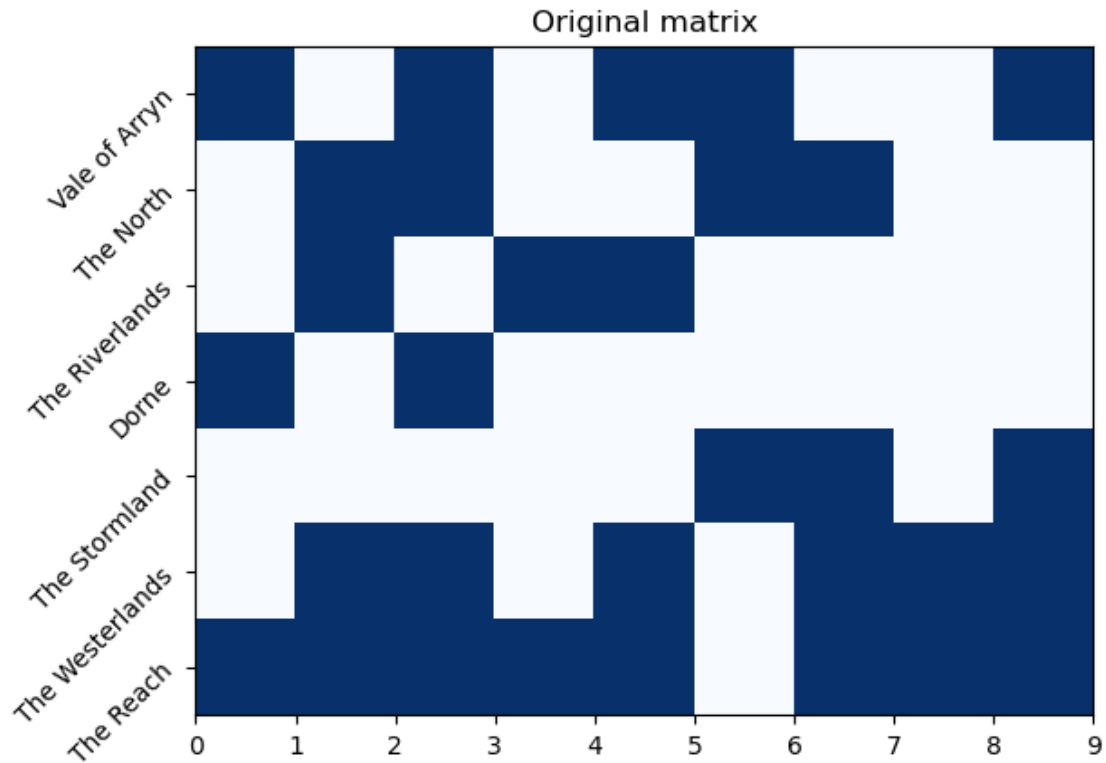
1.3.2 Visualize the network

```
[4]: Index1.show_graph()
```



1.3.3 Visualize the matrix

```
[5]: Index1.show_matrix()
```



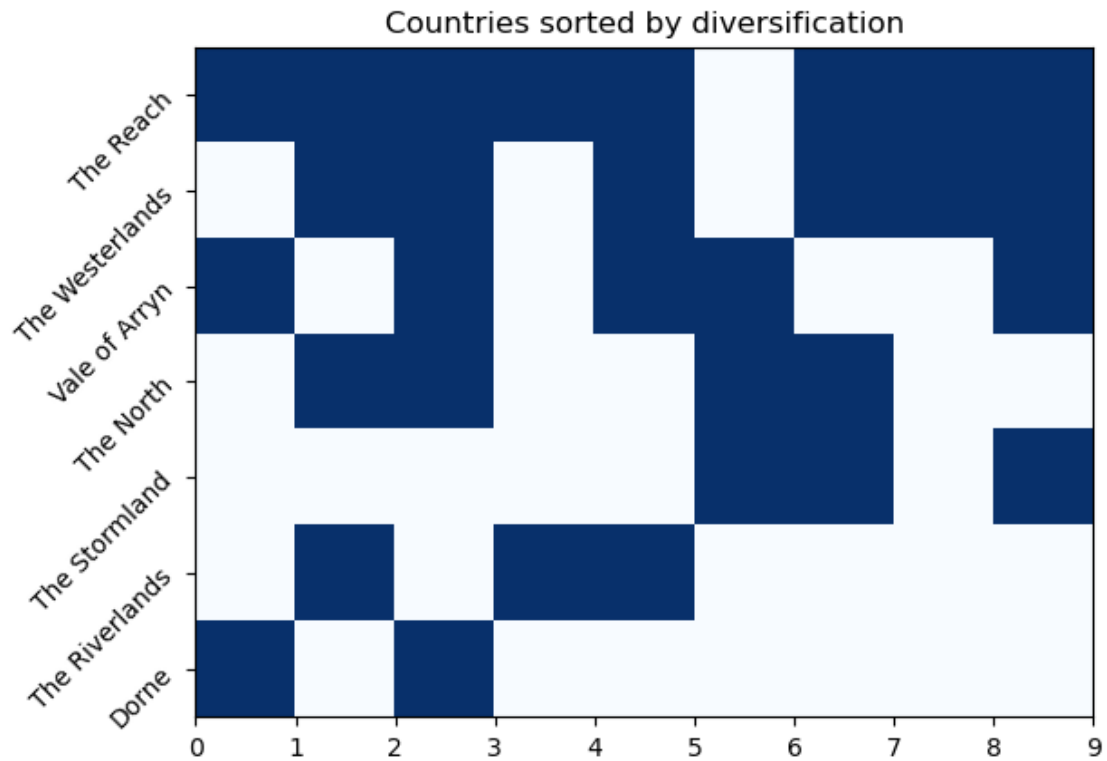
1.3.4 Assessing the complexity of countries

```
[6]: Index1.print_diversification()
```

Countries sorted by diversification (Most diverse at the top):

```
1. - The Reach          (diversification: 8)
2. - The Westerlands    (diversification: 6)
3. - Vale of Arryn      (diversification: 5)
4. - The North          (diversification: 4)
5. - The Stormland      (diversification: 3)
6. - The Riverlands     (diversification: 3)
7. - Dorne              (diversification: 2)
```

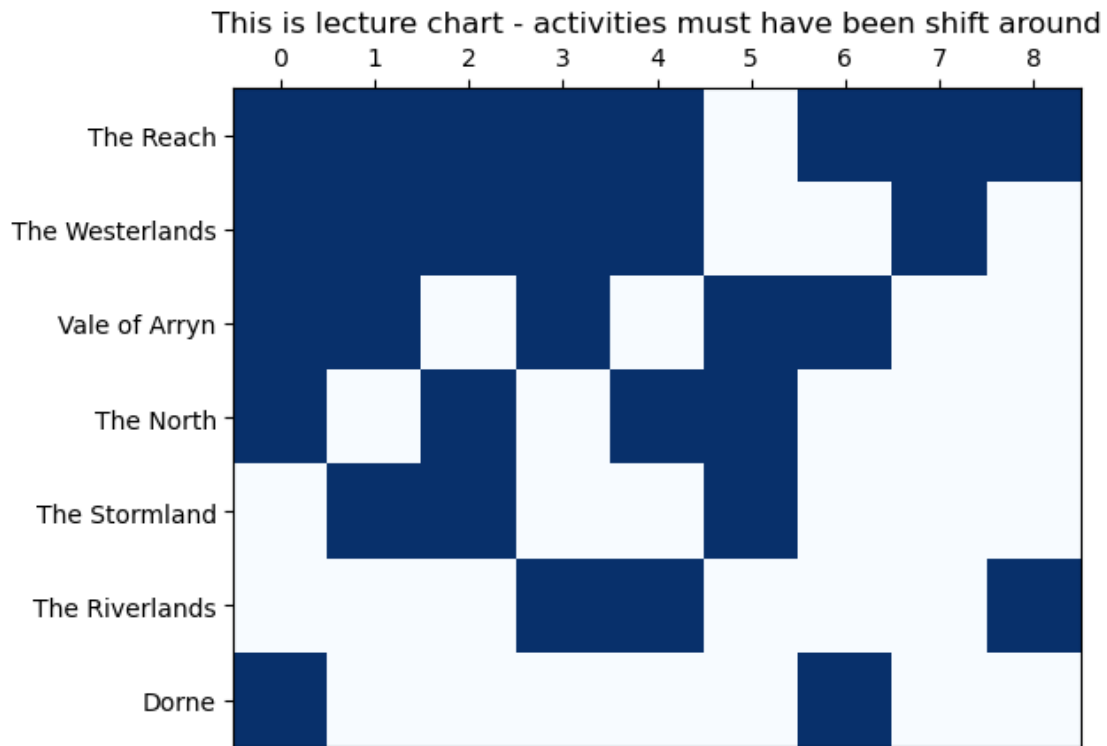
```
[7]: Index1.show_triangle()
```

1.3.5 Nestedness

Notice that Activity/Product Numbers do not match the actual numbers...

```
[8]: Index1.show_diversification()
```



1.3.6 Ubiquity

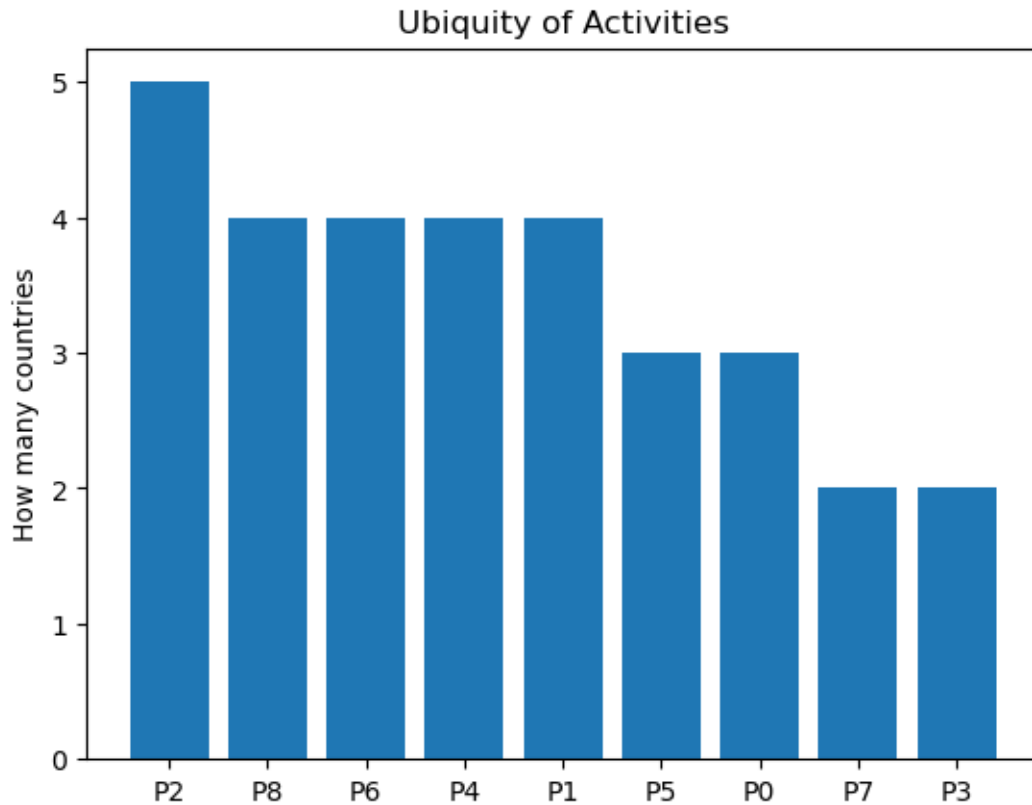
```
[9]: Index1.print_ubiquity()
```

Activities sorted by ubiquity (Most common at the top):

1. - P2 (ubiquity: 5)
2. - P8 (ubiquity: 4)
3. - P6 (ubiquity: 4)
4. - P4 (ubiquity: 4)
5. - P1 (ubiquity: 4)
6. - P5 (ubiquity: 3)
7. - P0 (ubiquity: 3)
8. - P7 (ubiquity: 2)
9. - P3 (ubiquity: 2)

--- --- ---

```
[10]: Index1.show_ubiquity()
```



1.3.7 Method of Reflections

- First one step only
- Then 10 iterations (even number)
- Then 19 iterations (odd number)

(For comparison)

```
[11]: Index1.do_method_of_reflections_once()
```

Method of Reflections (One Step only):

0: Vale of Arryn	= 3.8
1: The North	= 4.0
2: The Riverlands	= 3.3333
3: Dorne	= 4.0
4: The Stormland	= 3.6667
5: The Westerlands	= 3.8333
6: The Reach	= 3.5

--- --- ---

```
[12]: Index1.do_method_of_reflections()
```

```
Method of Reflections (Iterations: 10):
    0: Vale of Arryn      = -0.56652
    1: The North          = -0.50457
    2: The Riverlands     = 2.0324
    3: Dorne              = -0.67374
    4: The Stormland      = -1.1872
    5: The Westerlands    = 0.33608
    6: The Reach          = 0.56354
```

```
Ordered:
    1. - (2) The Riverlands      = 2.0324
    2. - (6) The Reach           = 0.56354
    3. - (5) The Westerlands     = 0.33608
    4. - (1) The North           = -0.50457
    5. - (0) Vale of Arryn       = -0.56652
    6. - (3) Dorne               = -0.67374
    7. - (4) The Stormland       = -1.1872
```

```
Ordered (sign removed):
    1. - (2) The Riverlands      = 2.0324
    2. - (4) The Stormland       = 1.1872
    3. - (3) Dorne               = 0.67374
    4. - (0) Vale of Arryn       = 0.56652
    5. - (6) The Reach           = 0.56354
    6. - (1) The North           = 0.50457
    7. - (5) The Westerlands     = 0.33608
```

```
[13]: Index1.do_method_of_reflections(19)
```

```
Method of Reflections (Iterations: 19):
    0: Vale of Arryn      = 0.57397
    1: The North          = 0.45588
    2: The Riverlands     = -2.0584
    3: Dorne              = 0.76801
    4: The Stormland      = 1.1159
    5: The Westerlands    = -0.31634
    6: The Reach          = -0.53896
```

```
Ordered:
    1. - (4) The Stormland      = 1.1159
    2. - (3) Dorne              = 0.76801
    3. - (0) Vale of Arryn      = 0.57397
    4. - (1) The North          = 0.45588
    5. - (5) The Westerlands    = -0.31634
    6. - (6) The Reach          = -0.53896
    7. - (2) The Riverlands     = -2.0584
```

```
Ordered (sign removed):
    1. - (2) The Riverlands     = 2.0584
```

```

2. - (4) The Stormland      = 1.1159
3. - (3) Dorne              = 0.76801
4. - (0) Vale of Arryn     = 0.57397
5. - (6) The Reach         = 0.53896
6. - (1) The North         = 0.45588
7. - (5) The Westerlands   = 0.31634

```

1.3.8 Chart ECI

Do with Method of Reflections and iterations 20

- with a sign
- without a sign

```
[14]: Index1.do_method_of_reflections(20)
```

Method of Reflections (Iterations: 20):

```

0: Vale of Arryn      = -0.55986
1: The North          = -0.48475
2: The Riverlands     = 2.054
3: Dorne              = -0.68683
4: The Stormland      = -1.1712
5: The Westerlands    = 0.30378
6: The Reach          = 0.54483

```

Ordered:

```

1. - (2) The Riverlands    = 2.054
2. - (6) The Reach         = 0.54483
3. - (5) The Westerlands   = 0.30378
4. - (1) The North        = -0.48475
5. - (0) Vale of Arryn    = -0.55986
6. - (3) Dorne            = -0.68683
7. - (4) The Stormland     = -1.1712

```

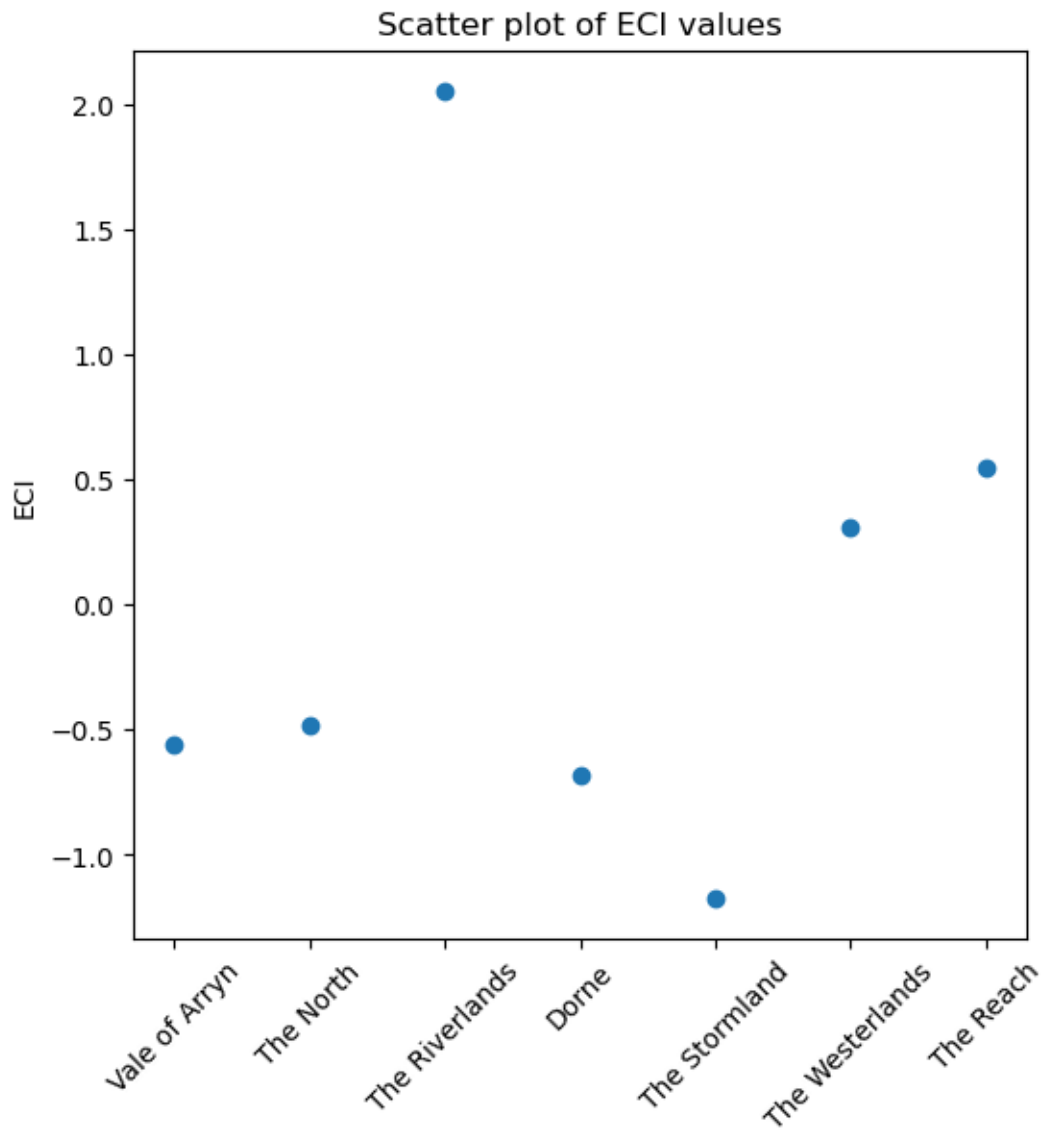
Ordered (sign removed):

```

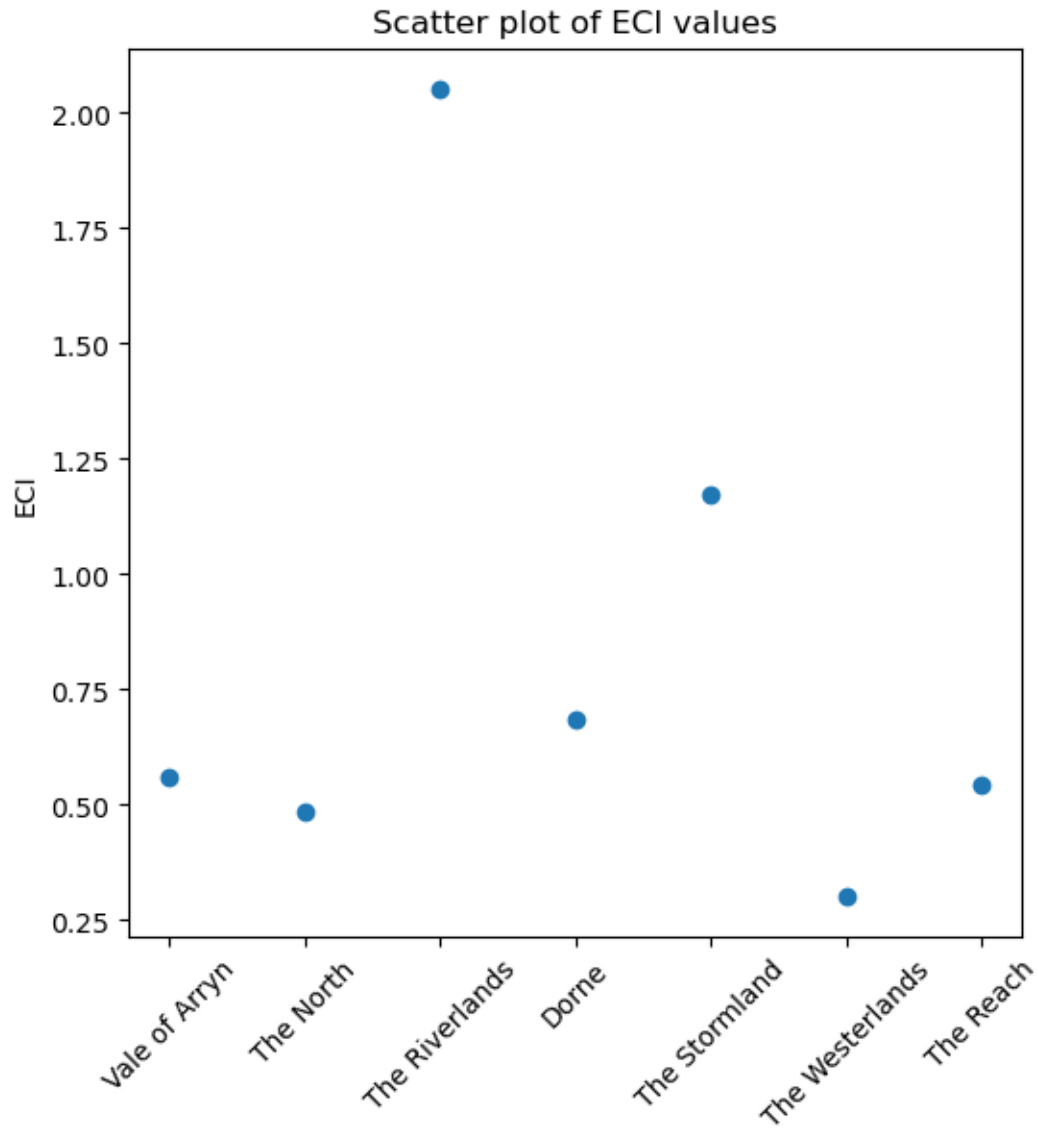
1. - (2) The Riverlands    = 2.054
2. - (4) The Stormland     = 1.1712
3. - (3) Dorne            = 0.68683
4. - (0) Vale of Arryn    = 0.55986
5. - (6) The Reach         = 0.54483
6. - (1) The North        = 0.48475
7. - (5) The Westerlands   = 0.30378

```

```
[15]: # keep sign
Index1.show_eci()
```



```
[16]: # removing sign and reordering  
Index1.show_eci(True)
```



1.3.9 ECI with eigenvalues

```
[17]: Index1.do_eigen_values_eci()
```

ECI as an eigenvalue problem:

0: Vale of Arryn	= -0.56136
1: The North	= -0.48151
2: The Riverlands	= 2.0548
3: Dorne	= -0.6958
4: The Stormland	= -1.1651
5: The Westerlands	= 0.30489

```

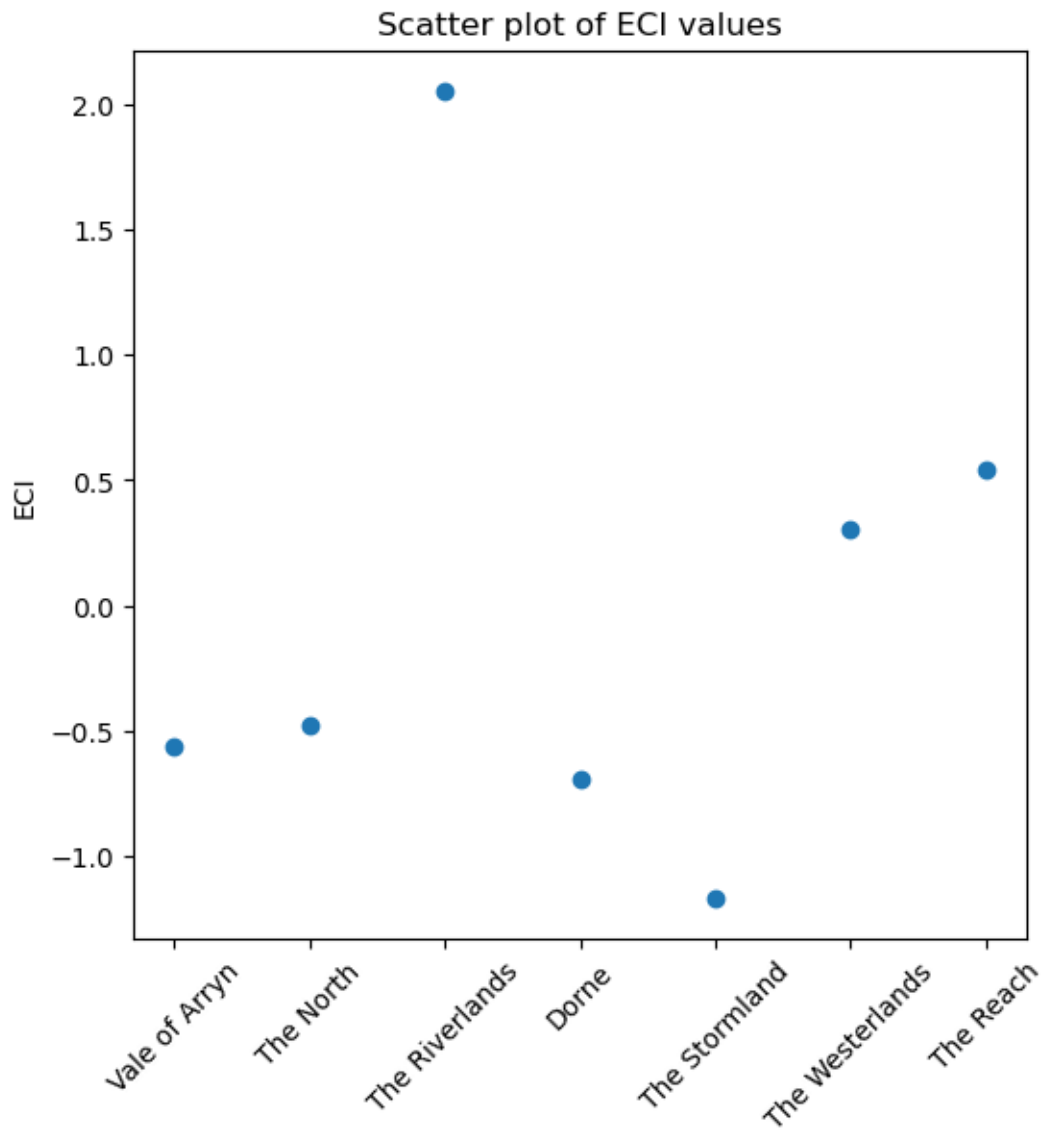
        6: The Reach                = 0.54408
---
Ordered:
    1. - (2) The Riverlands        = 2.0548
    2. - (6) The Reach              = 0.54408
    3. - (5) The Westerlands       = 0.30489
    4. - (1) The North              = -0.48151
    5. - (0) Vale of Arryn         = -0.56136
    6. - (3) Dorne                 = -0.6958
    7. - (4) The Stormland         = -1.1651
Ordered (sign removed):
    1. - (2) The Riverlands        = 2.0548
    2. - (4) The Stormland         = 1.1651
    3. - (3) Dorne                 = 0.6958
    4. - (0) Vale of Arryn         = 0.56136
    5. - (6) The Reach              = 0.54408
    6. - (1) The North              = 0.48151
    7. - (5) The Westerlands       = 0.30489
--- --- ---

```

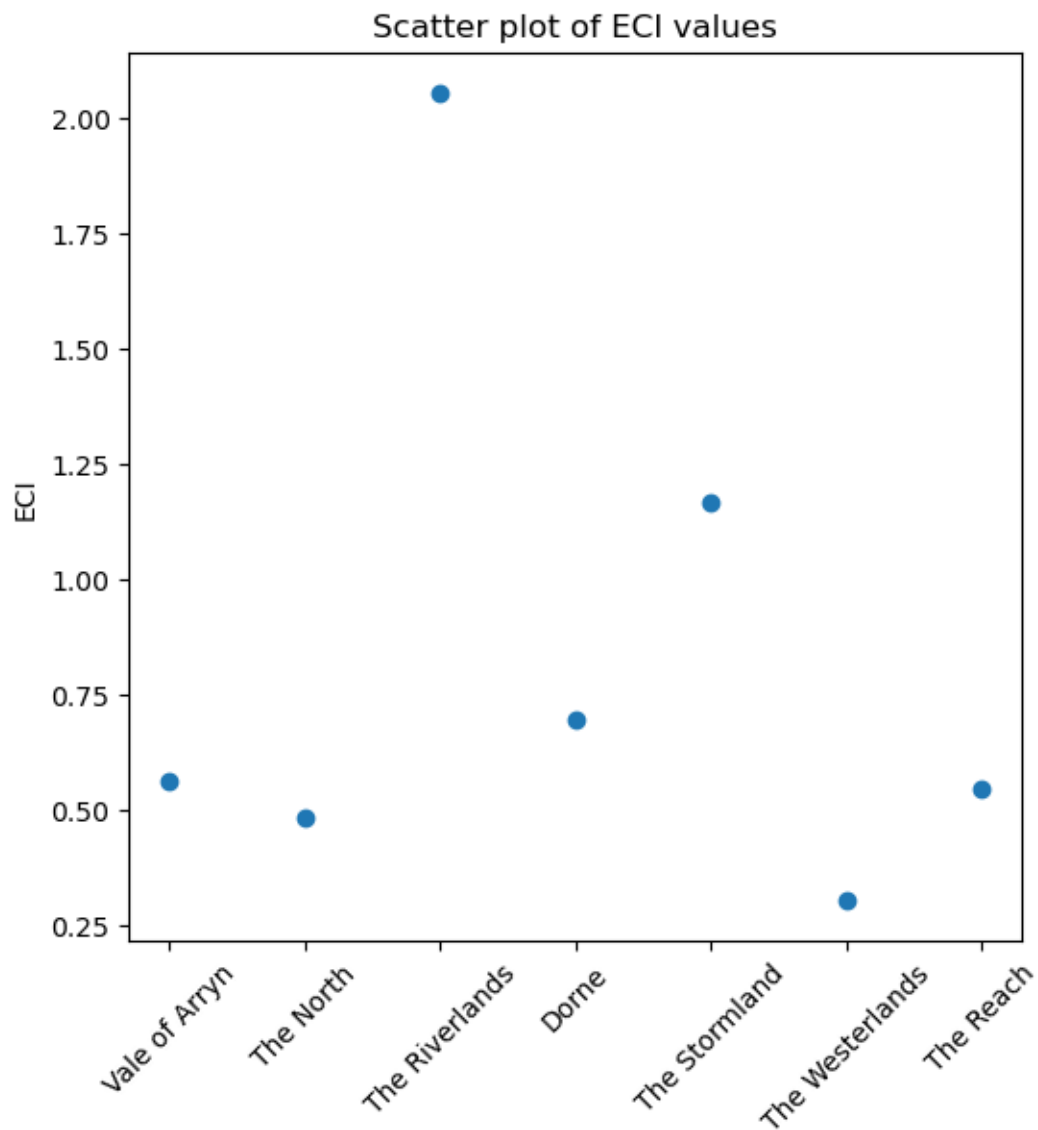
```

[18]: # keep sign
      Index1.show_eci()

```

```
[19]: # removing sign  
Index1.show_eci(True)
```



1.3.10 PCI with eigenvalues (Product Complexity Index)

```
[20]: Index1.do_eigen_values_pci()
```

PCI as an eigenvalue problem:

```
0: P0 = -0.66008
1: P1 = 0.77784
2: P2 = -0.55819
3: P3 = 1.961
4: P4 = 0.7438
5: P5 = -1.5098
```

```

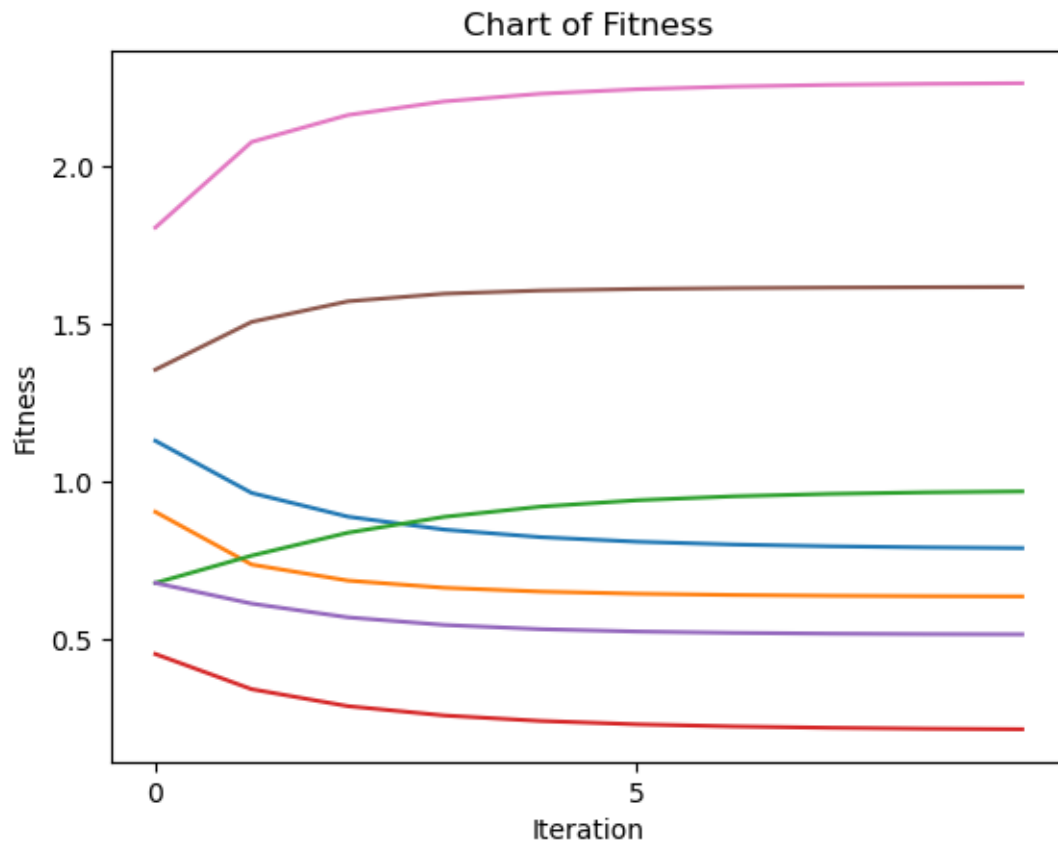
        6: P6 = -0.59482
        7: P7 = 0.46905
        8: P8 = -0.62886
---
Ordered:
    1. - (3) P3 = 1.961
    2. - (1) P1 = 0.77784
    3. - (4) P4 = 0.7438
    4. - (7) P7 = 0.46905
    5. - (2) P2 = -0.55819
    6. - (6) P6 = -0.59482
    7. - (8) P8 = -0.62886
    8. - (0) P0 = -0.66008
    9. - (5) P5 = -1.5098
Ordered (sign removed):
    1. - (3) P3 = 1.961
    2. - (5) P5 = 1.5098
    3. - (1) P1 = 0.77784
    4. - (4) P4 = 0.7438
    5. - (0) P0 = 0.66008
    6. - (8) P8 = 0.62886
    7. - (6) P6 = 0.59482
    8. - (2) P2 = 0.55819
    9. - (7) P7 = 0.46905
--- --- ---

```

1.4 Fitness and Complexity - another index of Complexity

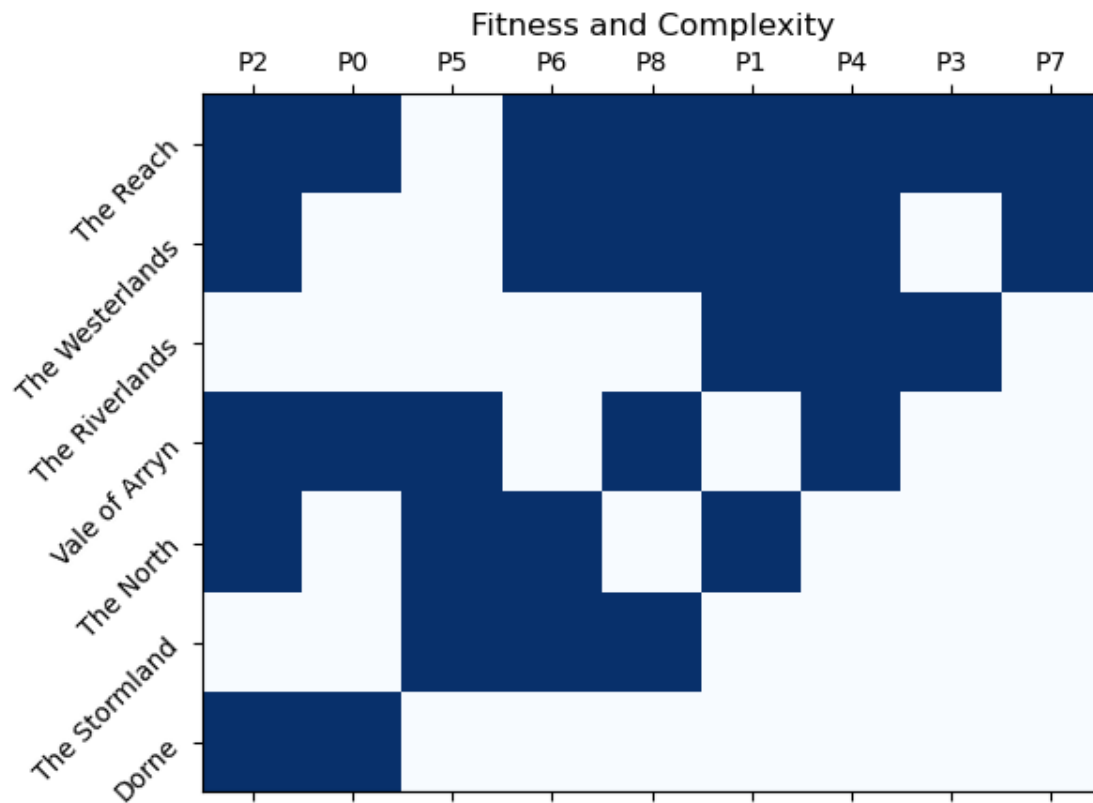
1.4.1 Fitness

```
[21]: Index1.show_fitness()
```



1.4.2 Fitness and Complexity

```
[22]: Index1.show_fitness_and_complexity()
```



1.5 Relatedness

1.5.1 Co-occurrences

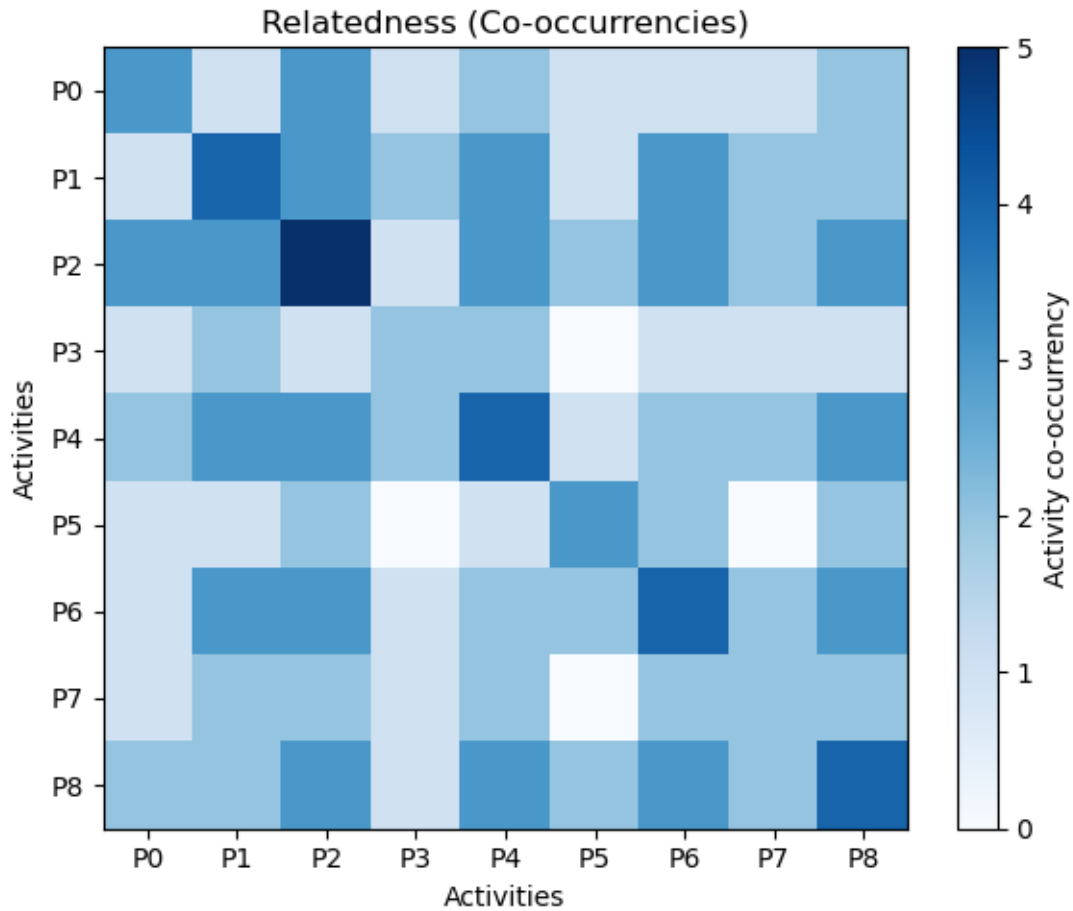
```
[23]: Index1.print_cooccurrences()
```

Relatedness / Co-occurrences

```
[[3 1 3 1 2 1 1 1 2]
 [1 4 3 2 3 1 3 2 2]
 [3 3 5 1 3 2 3 2 3]
 [1 2 1 2 2 0 1 1 1]
 [2 3 3 2 4 1 2 2 3]
 [1 1 2 0 1 3 2 0 2]
 [1 3 3 1 2 2 4 2 3]
 [1 2 2 1 2 0 2 2 2]
 [2 2 3 1 3 2 3 2 4]]
```

--- --- ---

```
[24]: Index1.show_activity_heatmap()
```



1.5.2 Relatedness (4 product space)

```
[25]: Index1.print_relatedness()
```

```
Relatedness / Product Space (4 products)
[[1.         0.25        0.6         0.33333333]
 [0.25        1.         0.6         0.5        ]
 [0.6         0.6         1.         0.2        ]
 [0.33333333 0.5         0.2         1.         ]]

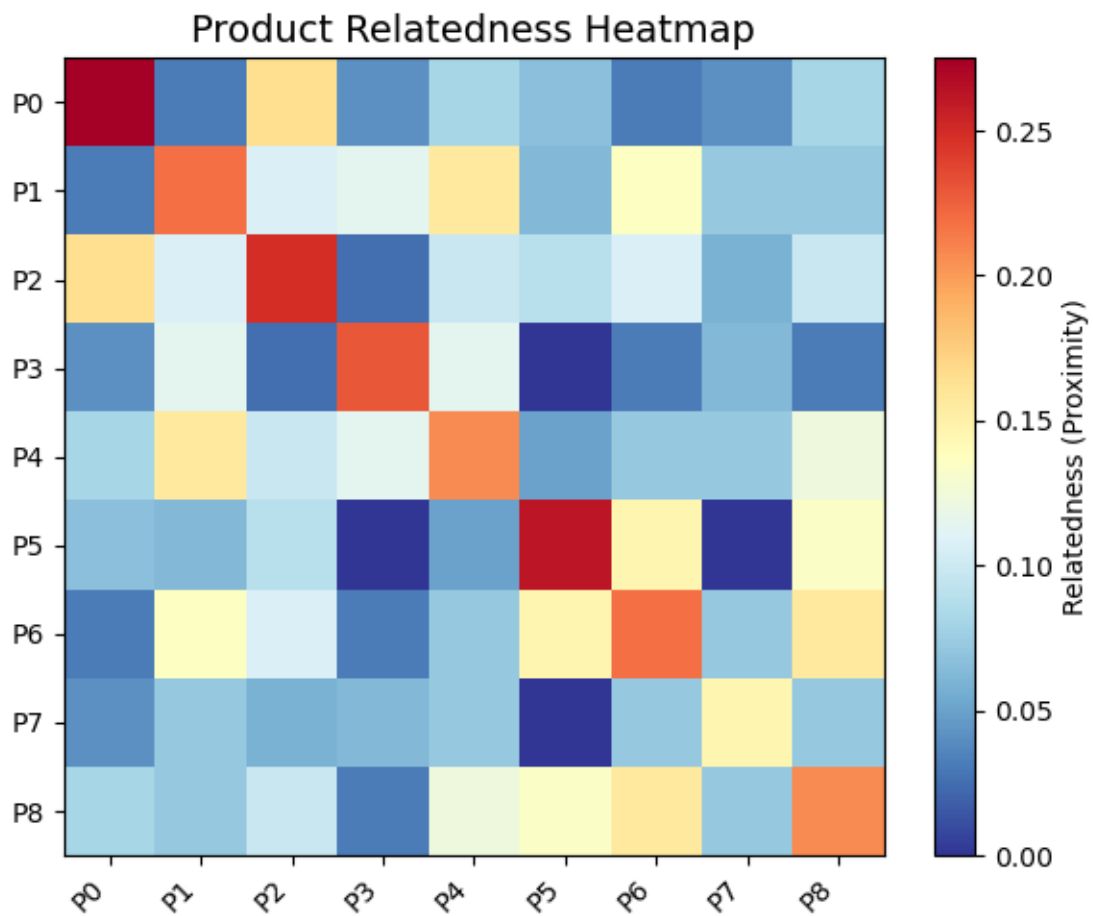
---
Relatedness / Taxonomy (4 products)
[[0.275      0.03125    0.165      0.04166667]
 [0.03125    0.21875    0.10833333 0.11458333]
 [0.165      0.10833333 0.24833333 0.025      ]
 [0.04166667 0.11458333 0.025      0.22916667]]

---
Relatedness / Assist Matrix (4 products)
```

```
[[0.275      0.03125   0.165     0.0625    ]
 [0.04166667 0.21875   0.10833333 0.22916667]
 [0.275      0.13541667 0.24833333 0.0625    ]
 [0.04166667 0.11458333 0.025     0.22916667]]
---
```

1.5.3 Proximity Heatmap

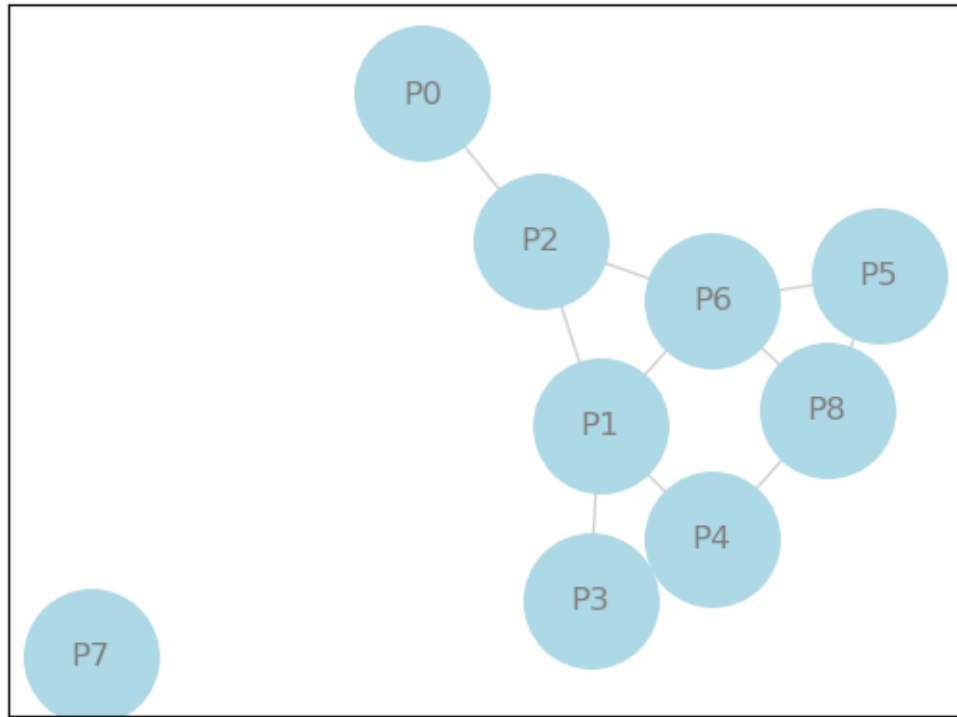
```
[26]: Index1.show_proximity_heatmap(6, 5)
```



1.5.4 Show Network of Taxonomy

Threshold 0.1

```
[28]: Index1.show_relatedness()
```



[]: